2016 Trump's Primary Elections Votes

and

US County Clusters

A Classification and Clustering Analysis

By

Rigos Anastasios

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Contents

1 Introduction

This report analyzes data from the presidential primary elections of 2016 in the United States in order to understand the socioeconomic characteristics of the US counties and how they relate to the voting. The analysis is divided into two parts, with the first one aiming at building a classification model to predict whether Donald Trump got more than 50% of the votes of Republican Party in a county, using the socioeconomic characteristics of the county as predictors. For this part of the analysis, the three different classification methods of Logistic Regression, Support Vector Machines with Radial Kernel and Random Forest are used, and their respective performance is also compared.

The second part of the analysis focuses on clustering the counties in the United States based on their demographic characteristics and trying to interpret and describe these clusters based on the economic related characteristics of the counties. Finally, an effort is made to interpret the clusters in terms of whether Trump received or not more than 50% of the votes in the Republican Party in the presidential primary elections of 2016.

The report is structured in seven different sections, Section 2 provides the Data overview and the necessary preprocessing steps for the analysis, including the merging process, missing values, and the variable groups. Section 3, in turn, presents the univariate exploratory data analysis, including the distribution of the response variable of the classification model and the key variables in the datasets. Section 4, next, examines the bivariate relationships between the predictors and the response variable for the classification analysis, as well as the correlations among the variables. Subsequently, Section 5 covers the classification analysis (Part 1), including the three methods used, their results, and a comparison of their performance and Section 6 presents the clustering analysis (Part 2), including the variable selection, the clustering results, the interpretation of the clusters using the economic variables, and the cluster validation. Finally, Section 7 concludes the report with a summary of the findings, while also connecting the two parts and trying to examine if in different clusters of counties, Trump got more or less than 50% of the votes of the Republicans.

2 Data Overview and Preprocessing

The data are provided in an Excel file containing two sheets and a data dictionary. The first sheet contains socioeconomic information for 3,195 counties in the United States, described by 54 variables. However, not all rows represent counties of the US, since the row with FIPS code equal to 0 corresponds to the whole United States, and 51 additional rows correspond to the 50 states and the District of Columbia, where the FIPS code ends in 000. Therefore, these 52 rows were removed from the dataset, resulting in a final dataset with 3,143 rows, each corresponding to a specific US county.

The 51 explanatory variables of the first sheet cover a wide range of characteristics related to the counties, including 22 demographic variables and 29 related to the economic conditions of the counties. The demographic group of the variables consist of population estimates (PST045214, PST040210, POP010210), population change (PST120214), age distribution (percentage under 5, under 18, and 65 and over), gender ratio, racial composition, percentage of people living in the same house, foreign-born population, veteran status, and two education variables (high school graduate and bachelor's degree or higher). The remaining 29 variables are related to the economy of the counties, such as household income, per capita income, poverty rates, housing units, homeownership rate, housing value, retail sales, manufacturing, and land area. This split between the 22 demographic and 29 economic variables is important for the clustering analysis in Section 6, where the demographic variables are used for clustering and the economic variables are used to describe the resulting clusters.

The second sheet contains 24,611 rows with votes from the 2016 presidential primary elections, with each row corresponding to the votes of a specific candidate in a specific county. The sheet includes votes from both the Democratic and Republican parties, but for the purpose of this analysis, only the votes for Donald Trump in the Republican primaries were used. After filtering for Trump's votes, 3,586 rows remained across different counties, however 10 of these records, all from New Hampshire, had missing FIPS codes. Since the county names were available from the first sheet, the correct FIPS codes were assigned manually by matching them with the first sheet (for example, Belknap County was mapped to FIPS 33001 and Hillsborough County to FIPS 33011). Following that, the response variable for the classification part of the analysis was created by assigning a value of 1 to counties where Trump received more than 50% of the votes and 0 otherwise.

An important consideration is that the data refer to the primary elections, which means that Trump's fraction of votes represents the share of votes he received among the Republican candidates, not the share of votes among all voters in the county. Additionally, the socioeconomic characteristics of the counties were collected between 20017 and 2014, while the votes are from 2016, which means that there is a small time gap between the predictors and the outcome of the classification analysis.

The two datasets were then merged based on the FIPS code, resulting in 422 counties being lost because votes were not available for them, because, maybe, in the counties that were not joined primaries did not take place. The counties that were not joined were from states, such as Kansas (105 counties), Minnesota (87), Colorado (64), and North Dakota (53). The final merged dataset for Classification analysis contains 2,721 rows and 56 variables. For Part 2, the full set of 3,143 counties was used for clustering, since the voting data are not needed.

3 Univariate Explanatory Data Analysis

Initial examination of the response variable of the classification part of the analysis shows that 1,724 counties (63.4%) had Trump receiving less than 50% of the votes, while 997 counties (36.6%) had Trump receiving more than 50% of the votes (Figure 1). Despite that the classes are not perfectly balanced, the imbalance is not severe, which means that the classification models should be able to perform well without the need for special techniques to handle class imbalance, such as oversampling or undersampling.

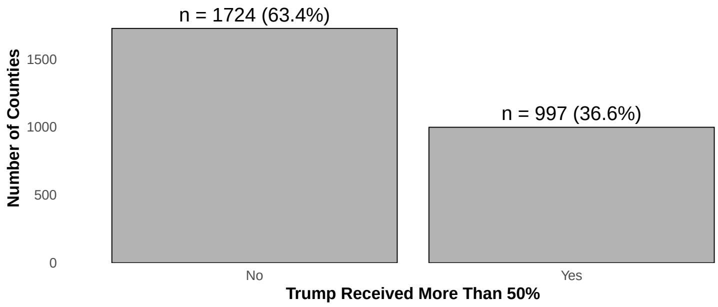

Figure 1: Barplot of the number of counties Trump received more than 50% ($N = 2{,}721$)

The density plot in Figure 2 shows the distribution of Trump's fraction of votes within the Republicans across the US counties. The peak of the distribution is around 0.45, while in most counties Trump received between 20% and 60% of the votes. Additionally, the number of countries of Trump receiving between 60% and 80% of the votes is approximately the same and in a very small number of counties, Trump received more than 80% of the votes. The dashed red line is showing the 50% threshold used to create the binary response variable for the classification part of the analysis. As can be seen in the plot, a large part of the distribution falls just below this threshold, which means that many counties were close to the boundary between the two classes.

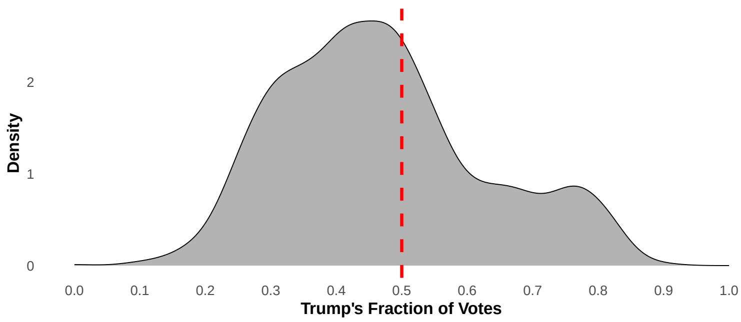

Figure 2: Density plot of Trump's fraction of votes in the Republican primaries across US counties ($N = 2{,}721$)

Having established the distribution of the response variable of the classification part, the analysis now continues with the explanatory variables, grouped into the five groups of population, race and ethnicity, education and age, income and poverty and housing variables.

Population

County populations ranged from 86 to over 10 millions (M = 104,307, SD = 329,063, Mdn = 27,496) and the distribution is extremely right-skewed (skewness = 14.32), as a small number of very large counties pull the mean upwards. For this reason, Figure 3 presents the distribution on a logarithmic scale(panel a). Population density showed a similar distribution (M = 270.5 people per square mile, SD = 1,819.4, Mdn = 49.7), with most counties having between 10 and 100 people per square mile, while a small number of highly urban counties reached over 10,000 (panel b). These patterns indicate that the majority of the counties in the dataset are small, while a few large counties have very high populations and population densities, which distorts the mean of the distributions.

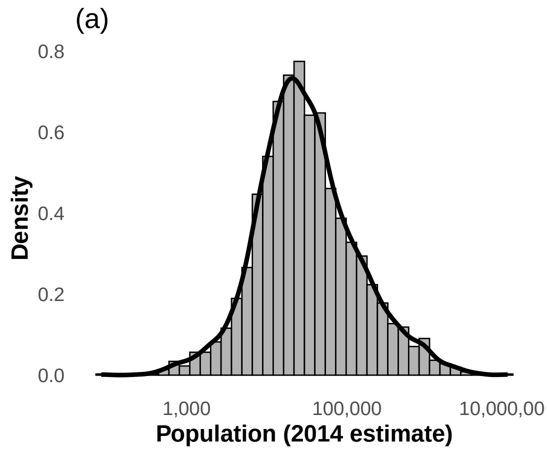

Figure 3: Distributions of population variables on a logarithmic scale (N = 2,721) (a) Population, 2014 estimate (b) Population per square mile.

Race and Ethnicity

Regarding the racial composition of the counties, in half of the counties, the percentage of White alone not Hispanic residents is above 83% (Mdn = 83.2%), and as can be seen from panel (a) the majority of counties have a high percentage of White residents. In contrast, the percentage of Black residents (M = 10.3%, SD = 15.1%, Mdn = 3.0%) and the percentage of Hispanic or Latino residents (M = 9.1%, SD = 13.9%, Mdn = 3.8%) were both heavily right-skewed (panels b and c), which indicates that most counties had very low percentages of Black and Hispanic residents, but some counties had much very high percentages. Similarly, most counties have less than 5% foreign-born residents (panel d).

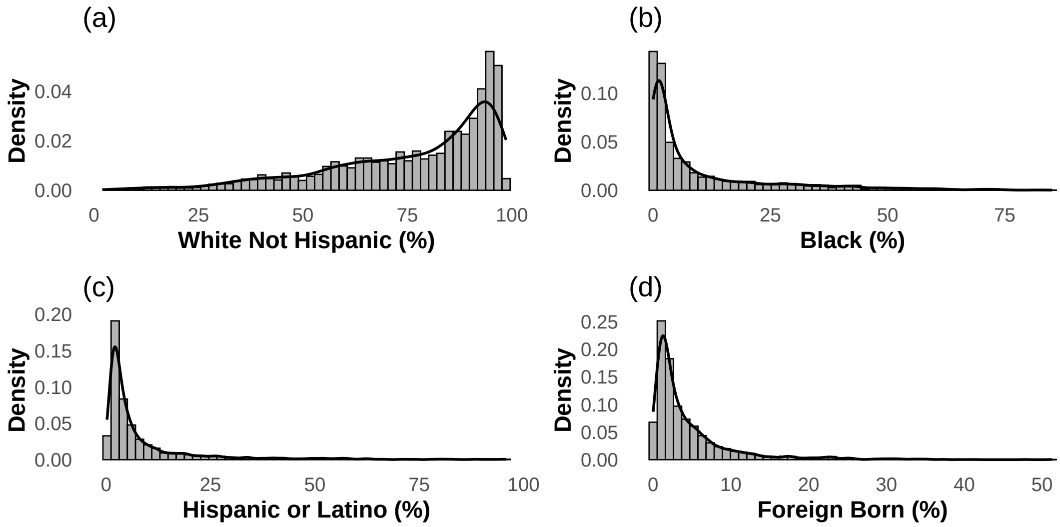

Figure 4: Distributions of race and ethnicity variables ($N = 2{,}721$) (a) White alone not Hispanic (b) Black (c) Hispanic or Latino (d) Foreign born

Education and Age

Turning to education and age (Figure 5), a clear difference emerges between the two levels of education. High school graduation is the standard across the US, since most counties have rates above 80% and the median is at 85.1% (M = 83.8%, SD = 6.9%). People with bachelor's degrees holders, however, vary much more across counties. Most counties have only 16.9% of its population aged 25 and over with a bachelor's degree or higher, and three quarters of counties are below 22.5% (M = 19.1%, SD = 8.5%). At the same time, a few counties reach above 40%, and the most educated county in the dataset reaches 74.4% (panel a). This gap between the two education levels suggests that while finishing high school is common everywhere, having a university degree differs a lot across the country.

Regarding the age distributions, the percentage of persons aged 65 and over (M = 17.6%, SD = 4.3%, Mdn = 17.2%) is centered around 17% (panel c), while the percentage of persons under 18 (M = 22.6%, SD = 3.3%, Mdn = 22.5%) is also approximately symmetric, with most counties having between 20% and 25% of their population under 18 (panel d).

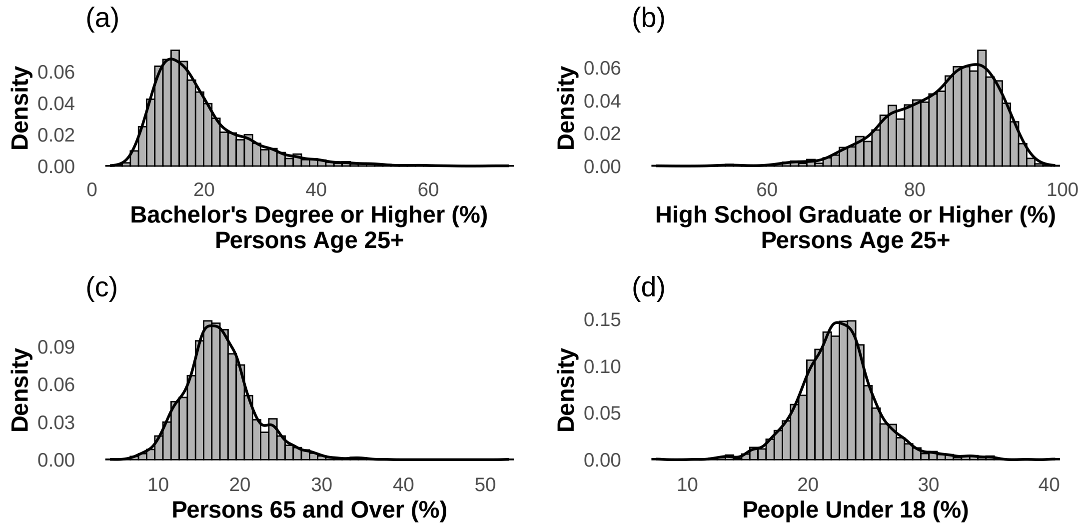

Figure 5: Distributions of education and age variables (N = 2,721) (a) Bachelor's degree or higher (%) (b) High school graduate or higher (%) (c) Persons 65 and over (%) (d) Persons under 18 (%)

Income and Poverty

Following the age and education variables, half of the counties earn less than \$43,273 per year, and in three out of four counties the income stays below \$50,102 (M = \$44,990, SD = \$11,705). The boxplot in panel (b) also shows that there are some wealthier countries, which are outliers. Additionally, the percentage of persons below the poverty level (M = 17.3%, SD = 6.5%, Mdn = 16.6%) is presented in the panels (c) and (d), The boxplot in panel (d) shows that the median poverty rate is 16.6%, with the middle 50% of counties falling between approximately 12% and 22%.

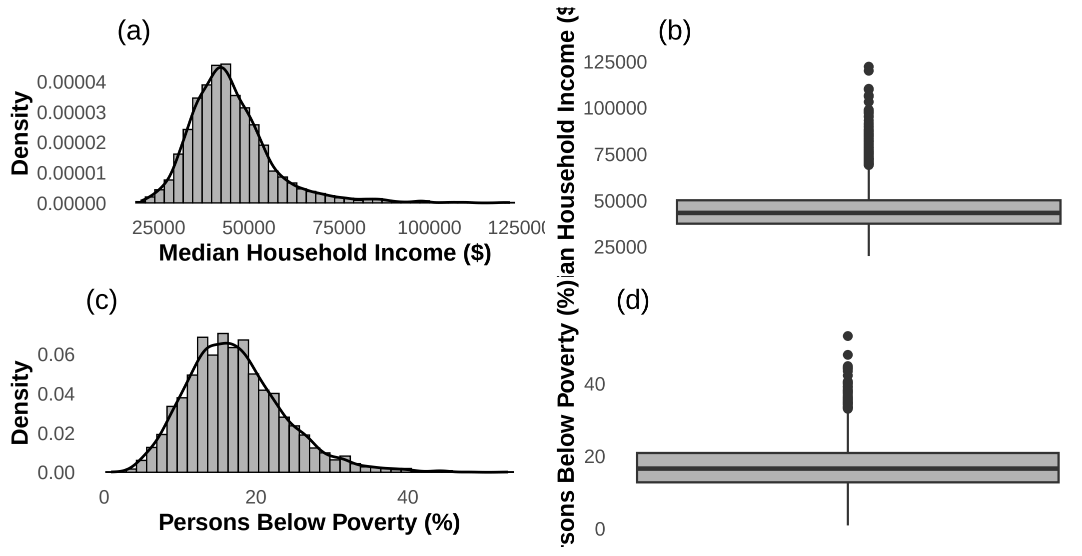

Figure 6: Distributions of income and poverty variables (N = 2,721) (a) Median household income histogram with KDE (b) Median household income boxplot (c) Persons below poverty level (%) histogram with KDE (d) Persons below poverty level (%) boxplot.

Housing

Finally, the housing variables were examined (Figure 7). Most counties in the dataset have high homeownership rates, while half of the counties have rates above 73.3% and the middle 50% falling between 68.2% and 77.5% (M = 72.1%, SD = 7.9%). However, a small number of counties fall well below 40%, as shown in the boxplot in panel (b), and these are most likely urban areas where renting is more common. Housing values tell a different story. While half of the counties have a median housing value below $106,800, the middle 50% ranges from $83,500 to $148,700, and some counties in expensive metropolitan areas exceed $500,000, with the most expensive county reaching $828,100. This creates a heavily right-skewed distribution (panels c and d), where the mean ($127,226) is pulled above the median by these expensive outliers. Overall, the US counties in the dataset are mostly places where people own their homes and housing is relatively affordable, but with a tail of expensive urban areas that stand out from the rest.

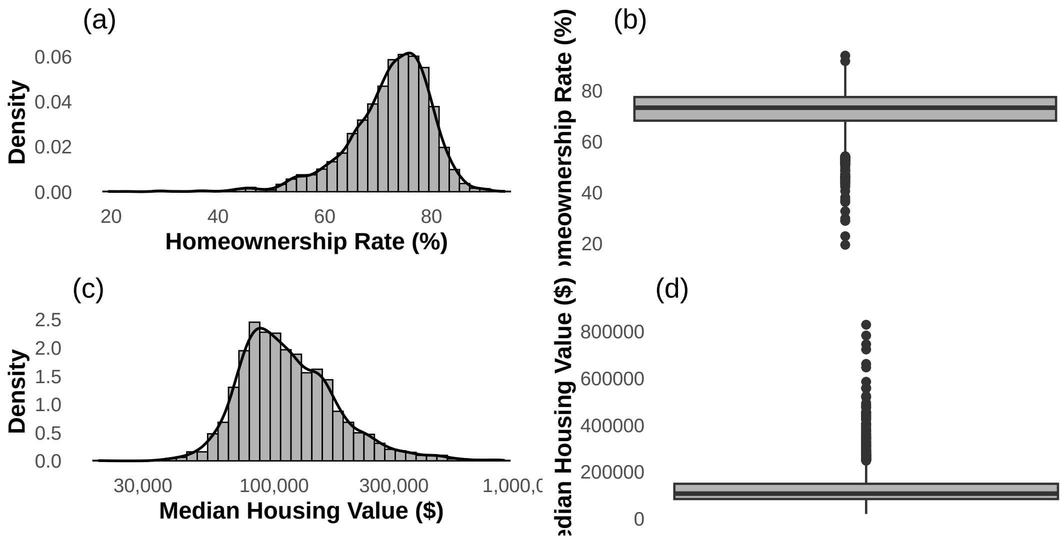

Figure 7: Distributions of housing variables (N = 2,721) (a) Homeownership rate (%) histogram with KDE (b) Homeownership rate (%) boxplot (c) Median housing value ($) histogram with KDE (d) Median housing value ($) boxplot.

4 Bivariate Exploratory Data Analysis

After the descriptive analysis of individual variables, examination of bivariate relationships was conducted in order not only to understand how each predictor is associated with whether Trump received more than 50% of the votes, but also how each variable is associated with each other.

Population

At first, the median population of the counties was significantly different between the two groups (Wilcoxon rank-sum test, W = 976,212, $p < .001$). Counties, where Trump got more than 50% of the votes had a median population of 22,864, which is considerably smaller than the median of 31,427 in counties that did not. The density plot in panel (a) shows that the grey density for counties that Trump did not have more than 50% is shifted to the right, meaning that larger counties were less likely to give Trump a majority. Population density also showed this pattern (W = 1,019,466, $p < .001$), as counties where Trump had the majority had a median of 39.5 people per square mile compared to 57.9. As can be seen in panel (b), the grey density again extends further to the right. Both plots use a logarithmic scale because the original distributions are extremely right-skewed. Overall, these results indicate that Trump performed better in smaller and more rural counties, while more populated and denser urban counties were less likely to give him a more than 50%.

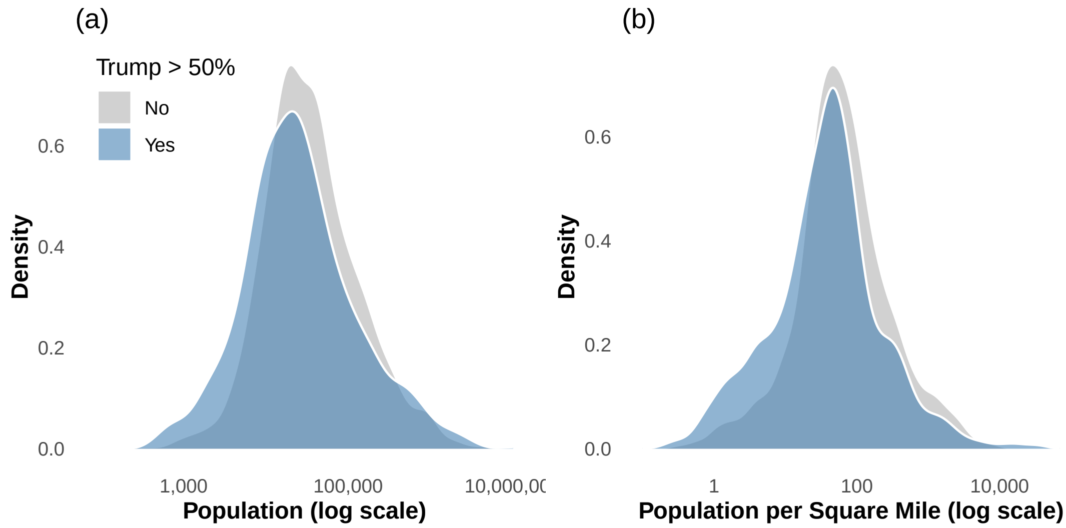

Figure 8: Density plots of population variables by Trump majority on a logarithmic scale (N = 2,721) (a) Population, 2014 estimate (b) Population per square mile.

Race and Ethnicity

Subsequently, racial and ethnic composition of the counties were also different between the two groups (Figure 9). Based on the univariate analysis and the presence of outliers in these distributions, the mean is not a sufficient measure of central location and consequently Wilcoxon rank-sum tests were used to compare the medians of the two groups. The percentage of White alone not Hispanic residents was higher in counties where Trump received more than 50% of the votes (W = 751,208, p < .001), with a median of 86.5% compared to 80.6% in the counties that did not. As shown in panel (a), the blue density for the counties that Trump had the majority of votes is shifted to the right compared to the grey density for counties that Trump did not, indicating that most counties where Trump got more that 50% have very high percentages of White residents. In contrast, the percentage of Black residents was lower in counties that Trump had more than 50% of the votes (W = 1,040,292, p < .001), with a median of 1.7% compared to 4.3%, as shown in panel (b) where the grey density is shifted more to the right.

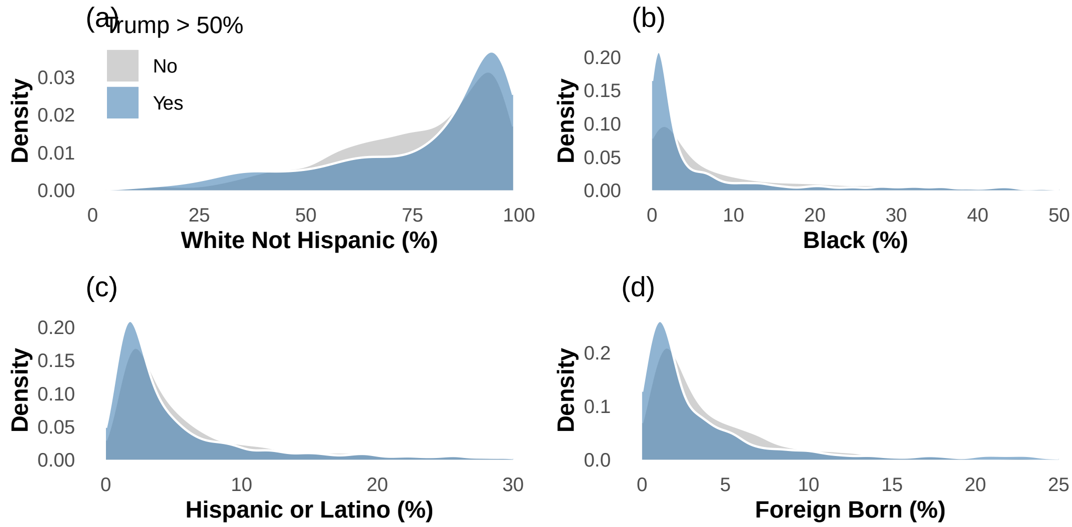

Figure 9: Density plots of race and ethnicity variables by Trump majority (N = 2,721) (a) White alone not Hispanic (%) (b) Black (%) (c) Hispanic or Latino (%) (d) Foreign born (%)

A similar pattern was found for the percentage of Hispanic or Latino residents (W = 984,605, p < .001). Countries where Trump had the majority of votes had a median of 3.1% in comparison of 4.2% in counties that he did not. The grey density in panel (c) is again shifted to the right, though the difference is smaller than for Black residents. Furthermore, the percentage of residents that were foreign born was also lower in counties that Trump received more that 50% of the votes (W = 995,104, p < .001), having a median of 1.9% compared to 2.8%. These findings indicate that Trump got the more than 50% of the votes in counties with less diverse populations, where the majority of residents were White and where there were less foreign-born residents.

Education and Age

Turning now to education and age, Figure 10 presents the density plots of four variables. The percentage of residents with a bachelor's degree or higher was examined using a Wilcoxon rank-sum test (right-skewed distribution), showing that counties where Trump did not receive a majority had a higher median of bachelor's degree holders (Mdn = 18.1%) compared to the counties Trump had more that 50% of the votes (Mdn = 15.5%). This difference was statistically significant (W = 1,297,864, p < .001) and it might suggest that more educated counties were less likely to vote for Trump in the primary elections.

Regarding high school graduation, no statistically significant difference was found between the two groups ($t(1967.6) = -1.09$, $p = .274$), with the mean percentage of high school graduates was nearly identical in both groups (M = 83.7% for counties Trump received less than 50% of the votes and M = 84.0% for counties Trump received more than 50% of the votes). However, as can be seen from the plot in panel (b) there is a difference in counties with 80% or more high school graduates. In conclusion, this indicates that while higher education differs between the groups, high school education does not.

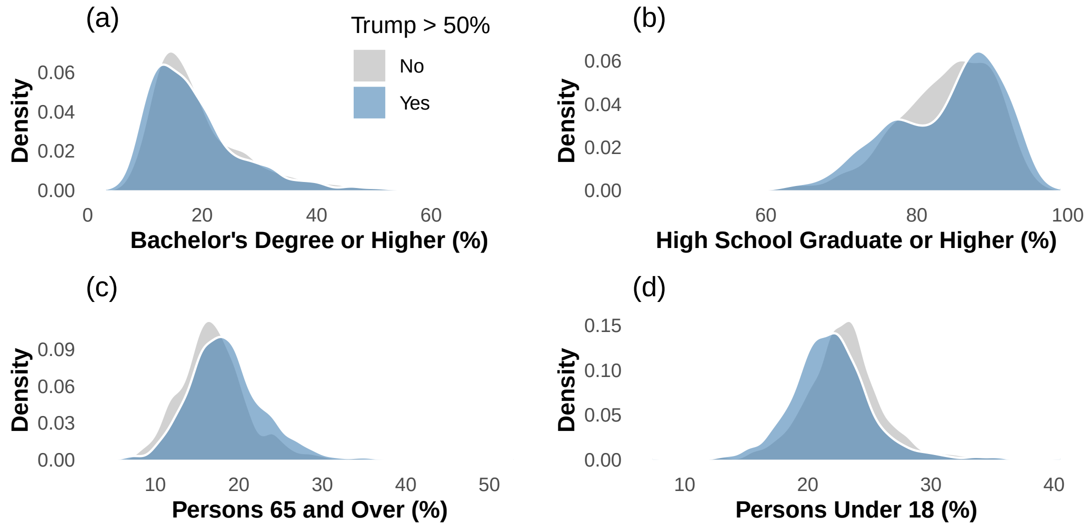

Figure 10: Density plots of education and age variables by Trump majority (N = 2,721) (a) Bachelor's degree or higher (%) (b) High school graduate or higher (%) (c) Persons 65 and over (%) (d) Persons under 18 (%).

Finally, the percentage of people aged 65 and over was significantly higher in counties that Trump received a majority of votes ($t(1905.3) = -8.36$, $p < .001$), with a mean of 18.5% compared to 17.0% in counties with less than 50% of the votes, which is also shown in panel (c), as the blue density is shifted slightly to the right, meaning that counties with older populations tended to give Trump more votes. In contrast, the percentage of persons under 18 showed the opposite pattern ($t(1973.5) = 7.52$, $p < .001$), with counties that Trump had less than 50% of the votes having a higher mean (M = 22.9%) than the other counties (M = 21.9%). These results suggest that Trump performed better in counties with older populations and lower levels of higher education.

Income and Poverty

Median household income was statistically different between the two groups (Wilcoxon rank-sum test, W = 942,174, $p < .001$), though the actual difference in medians was small. Counties where Trump had more than 50% of the voted had a median household income of $43,815 compared to $42,445 in counties that did not. The boxplots in panel (a) confirm that the difference is minor, as the two boxes overlap almost entirely and the median lines are very close to each other. However, the non-majority group has more outliers at the upper end, indicating that a number of wealthier counties did not give Trump a majority. While the difference is statistically significant, these two distributions are very similar.

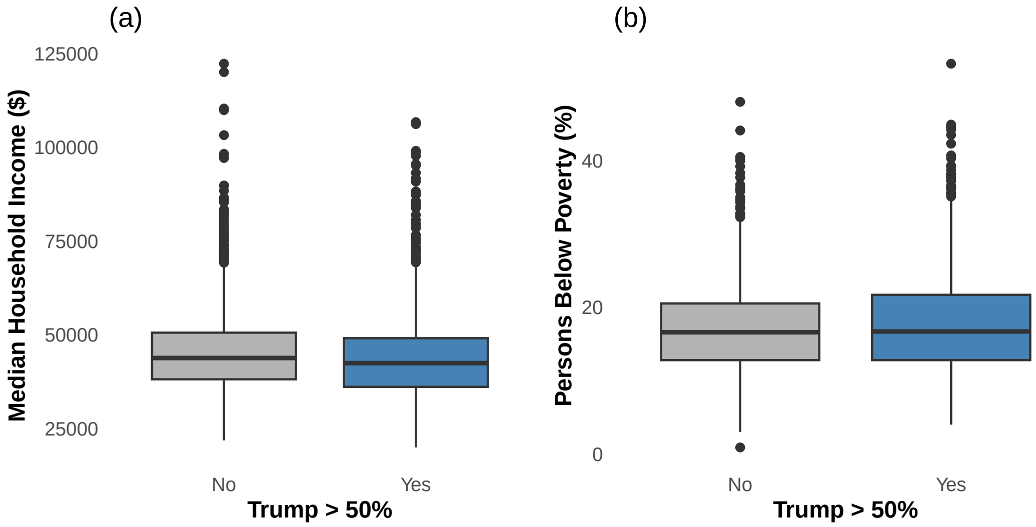

Figure 11: Boxplots of income and poverty variables by Trump majority (N = 2,721) (a) Median household income ($) (b) Persons below poverty level (%).

Regarding the percentage of persons below the poverty level, no statistically significant difference was found between the two groups (Wilcoxon rank-sum test, W = 831,458, p = .157). Both groups had a nearly identical median poverty rate (Mdn = 16.6% for more than 50% counties and Mdn = 16.7% for less than 50% counties). The boxplots in panel (b) show that the boxes are almost the same in size and position, with similar interquartile ranges and median lines at the same level. Both groups also have a number of outliers at the upper end, representing counties with very high poverty rates. This is a notable finding, as it suggests that poverty on its own is not associated with whether Trump received more than 50% of the votes. It is possible that poverty interacts with other variables such as racial composition or education level, but as a standalone predictor it does not distinguish between the two groups.

Housing

Finally, the housing variables were examined (Figure 12), in which the homeownership rate did not differ significantly between the two groups ($t(2026) = 1.10$, p = .273) and the boxplots in panel (a) show that both groups have similar medians and distributions. Similarly, median housing value was also not significantly different between the two groups (Wilcoxon rank-sum test, W = 882,358, p = .245), while the density plot in panel (b) shows that the grey density for counties that Trump did not receive more than 50% of the votes extends slightly further to the right, but the overall distributions are very similar. In conclusion, these results suggest that housing characteristics are not strongly associated with whether Trump received more than 50% of the votes.

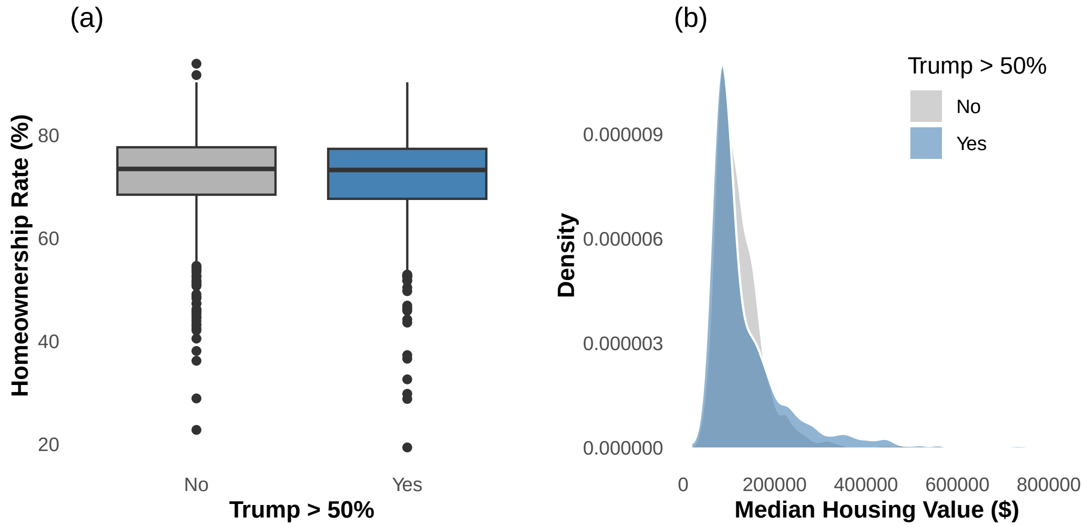

Figure 12: Housing variables by Trump majority (N = 2,721) (a) Homeownership rate (%) boxplot (b) Median housing value ($) density plot.

Associations Between Variables

After examining how key predictors relates to the response variable of the binary classification, Spearman correlations were computed between all 51 predictor variables to identify pairs that have similar information. In total, 289 pairs had $|r| > .50$, with 33 among the demographic variables, 138 among the economic variables, and 118 between the two groups.

The most important insight from these correlations is that many variables simply measure how large each county is. For example, the three population variables (population 2014, population 2010 estimates base, and population 2010 census) are all correlated with each other with $r =$ 1.00, and they are also nearly perfectly correlated with housing units, households, number of businesses, number of firms, retail sales ($r > .95$), and the number of veterans ($r = .98$). This means that in a large county, everything is larger in absolute terms, and all these variables carry the same information. This is important for both parts of the analysis, because including multiple variables related to country size would add noise not only in the classification models, but also in clustering.

Within the demographic variables, several other patterns stand out. The racial composition variables approximately sum to 100%, which creates strong negative correlations between the percentage of White residents and the percentages of Black ($r = -.84$) and Hispanic ($r = -.63$) residents. The variables related to immigration also move together, with the percentage of foreign-born persons being strongly correlated with the percentage of people who speak a language other than English at home ($r = .87$) and with the percentage of Hispanic residents ($r = .85$). Regarding age, the percentage of persons under 5 is included in the percentage of those under 18 ($r = .83$), and younger counties tend to have fewer elderly residents ($r = -.57$ between under 18 and 65 and over). Finally, the two education variables are positively correlated

(r = .69), though they measure different levels of education. Table 1 summarizes these key correlations.

Table 1: Key Spearman correlations among the demographic variables (N = 2,721).

Variable 1	Variable 2	r
Population 2014	Population 2010 (both)	1.00
Population 2014	Veterans (count)	.98
Foreign born (%)	Non-English at home (%)	.87
White alone (%)	White not Hispanic (%)	.85
Hispanic (%)	Foreign born (%)	.85
White alone (%)	Black (%)	−.84
Under 5 (%)	Under 18 (%)	.83
High school (%)	Bachelor's degree (%)	.69
White not Hispanic (%)	Hispanic (%)	−.63
Under 18 (%)	65 and over (%)	−.57

Among the economic variables, housing units, households, businesses, employment, retail sales, and accommodation sales are all correlated at $r > .90$ with each other. Additionally, there is a correlation between median household income and per capita income ($r = .86$), which are two different ways of measuring wealth, and between income and poverty ($r = -.83$), which shows that wealthier counties have lower poverty rates. Moreover, homeownership rate is negatively correlated with multi-unit housing ($r = -.63$), which makes sense since renters tend to live in apartment buildings.

Regarding the correlation between the economic and demographic variables, the demographic population variables are correlated at $r > .95$ with the economic count variables (housing units, businesses, retail sales). Beyond this, one also important correlation the one between education and income, in which the percentage of bachelor's degree holders is correlated with per capita income ($r = .75$) and with median household income ($r = .65$), while high school graduation is correlated with per capita income ($r = .73$) and negatively correlated with poverty ($r = -.67$). These associations may indicate that counties with more educated populations tend to be wealthier and have lower poverty rates.

For the classification models in Part 1, the handling of correlated variables was left to each method's own variable selection mechanism. For the clustering in Part 2, the redundant variables need to be removed before fitting the model, as will be discussed in Section 6.

5 Classification Analysis: Predicting Trump's Majority of Republican votes

The first part of the analysis aims to predict whether Donald Trump received more than 50% of the Republican primary votes in each county. Three classification methods were used for this purpose, each with a different approach to handling the large number of predictors. At first, Logistic regression with backward stepwise selection was used, then Support Vector Machine with Radial Kernel, and Random Forest which is an ensemble method. All three methods were trained and evaluated in the same training and test datasets using the cross-validation procedure described below.

Splitting Strategy

In order to train and evaluate the classification models, a supervised 5-fold cross-validation was repeated two times, which is also called as 2 times Repeated Stratified 5-fold Cross Validation. In this approach, the dataset is randomly divided into 5 equally sized folds, and in each iteration 4 folds are used for training while the remaining fold is used for testing. This process is repeated until every fold has served as the test set exactly once. This whole procedure is then repeated one more time, in order to have accurate and true estimates and this results in 10 folds in total.

One key observation about the training and the evaluation of the models is that all three models are trained in exactly the same training set and evaluated in the same set in each fold, in order to be able to compare directly the performance of each model with the other ones and also minimize the splitting bias.

Supervised Cross-Validation was used because the two classes are not perfectly balanced, as in 63.4% of the counties Trump did not receive the majority of the votes. Although this imbalance is not severe for oversampling or undersampling techniques, supervised sampling ensures that each fold has the same class proportions as the full dataset.

All 51 predictor variables were included in the modelling process, but the identifier variables (FIPS code, county name, state abbreviation) and the continuous fraction of votes were excluded. The handling of redundant and correlated variables was left to each method's own variable selection mechanism, as discussed in the respective subsections below.

Method 1: Logistic Regression

The first method used was logistic regression, which models the log-odds of the response variable as a linear combination of the predictors. Logistic regression was chosen because it is a well-established method for binary classification that provides interpretable coefficients and allows for variable selection.

In each fold, a full logistic regression model was first fitted using all 51 predictors and, subsequently, backward stepwise selection based on the Akaike Information Criterion was applied to remove variables that did not contribute meaningfully to the model. This variable selection was performed inside each fold separately, and in turn, the selected model was then used to predict the test fold, with a classification threshold of 0.50.

Across the 10 folds, logistic regression achieved a mean accuracy of 71.3% ($SD = 0.020$), a mean sensitivity of 45.4%, a mean specificity of 86.2%, and a mean precision of 65.8%. The relatively low sensitivity indicates that the model is better at identifying counties where Trump did not receive a majority than at identifying those where he did. This is expected given the class imbalance, as the model tends to be more conservative in predicting the minority class.

Regarding variable selection, 25 out of the 51 variables were selected in all 10 folds, indicating that they are stable and important predictors, including racial composition variables (White alone, Black, two or more races, Hispanic, White not Hispanic), education (bachelor's degree), income (per capita income), poverty, age (under 5, under 18), housing (homeownership rate, median housing value, multi-unit structures), business activity (establishments, employment), and several others. On the other hand, four variables of Native Hawaiian percentage, persons per household, Asian-owned firms percentage, and accommodation sales were never selected by the stepwise procedure, suggesting that these variables do not carry additional predictive information beyond what is already captured by the other predictors.

Method 2: Support Vector Machines

The second method used was a Support Vector Machines (SVM) with a radial kernel, and it was chosen because it can capture non-linear relationships between the predictors and the response variable through the kernel function, as it projects the data in a higher dimension to make them separable.

In each fold, a small grid search was done for the tuning of the cost parameter (values $\{1, 10\}$), which controls the trade-off between a wide margin and correct classification of training points, and the gamma parameter, which controls how far the influence of a single training example reaches, was searched over $\{0.01, 0.1\}$. In 9 out of 10 folds, the best parameters were cost $= 10$ and gamma $= 0.01$, while in one fold cost $= 1$ and gamma $= 0.1$ was selected, resulting in the lowest cross-validation error.

Support Vector Machine had a mean accuracy of 74.8% ($SD = 0.016$), a mean sensitivity of 47.7%, a mean specificity of 90.4%, and a mean precision of 74.4% in the 10 folds. The high specificity (90.4%) means that the model is very good at correctly identifying counties where Trump did not receive the majority, however it still misses more than half of the counties where he did. One limitation of SVM compared to logistic regression is that it is a 'black box' method and does not provide direct variable importance measures, which makes it harder to interpret which predictors drive the predictions.

Method 3: Random Forest

The third method used was Random Forest, which is an ensemble method that builds a large number of decision trees, each trained on a bootstrap sample of the data with a random subset of predictors considered at each split and the final prediction is obtained by majority voting across all trees. Random Forest was chosen because it is robust to correlated predictors since each tree only sees a random subset of them.

In each fold, a grid search was used over the number of variables considered at each split ($mtry \in \{5, 7, 10\}$) and the number of trees ($ntree \in \{100, 300, 500\}$), and in turn the combination with the lowest out-of-bag (OOB) error was selected. Across the 10 folds, $ntree = 500$ was selected in 7 folds and $mtry = 7$ was the most frequently selected value, appearing in 5 folds.

Random Forest achieved the highest mean accuracy of 77.6% ($SD = 0.012$), a mean sensitivity of 56.1%, a mean specificity of 90.1%, and a mean precision of 76.8%. The variable importance was assessed using the two metrics of Mean Decrease in Gini and Mean Decrease in Accuracy averaged across the 10 folds. Figure 13 shows the top 15 variables for each measure, and as a result the top five most important variables according to Mean Decrease Gini were land area, median housing value, persons under 18, Hispanic percentage, and Black percentage. The Mean Decrease Accuracy ranking is similar, with land area and median housing value again at the top, followed by Hispanic percentage, persons under 18, and Black percentage. These findings are consistent with the bivariate analysis, which showed that racial composition, age, and county size were the most strongly associated with the response variable.

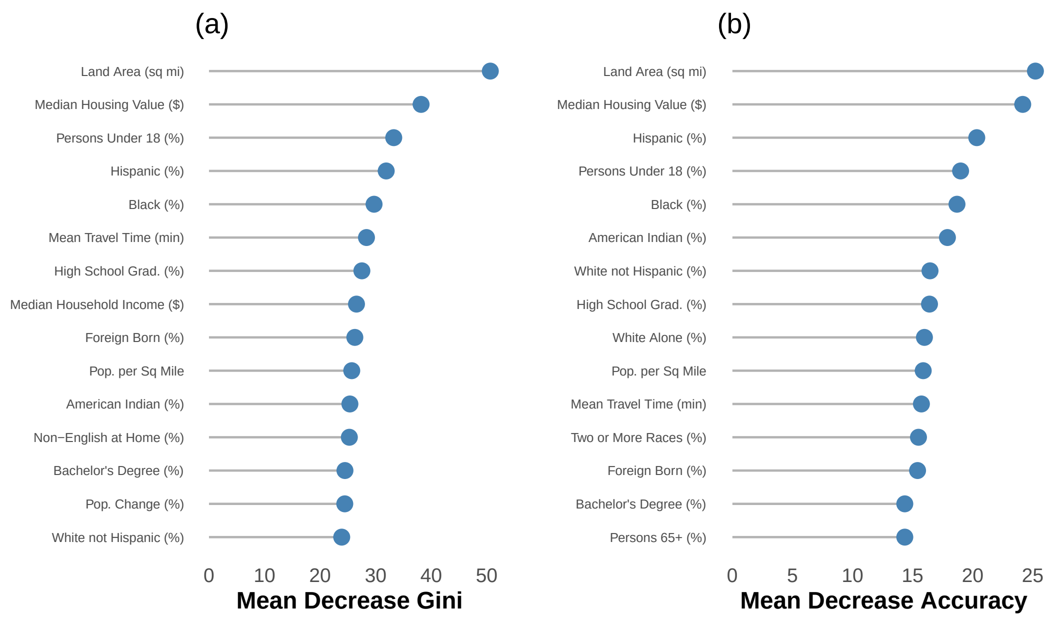

Figure 13: Random Forest variable importance averaged across 10 folds ($N = 2{,}721$) (a) Mean Decrease Gini (b) Mean Decrease Accuracy.

Model Comparison

Before comparing the three methods, it is useful to explain the different performance metrics used. Accuracy measures the overall percentage of correct predictions, however, when the classes are not balanced, accuracy alone can be misleading. Sensitivity (also known as recall) measures the percentage of counties where Trump actually received a majority that the model correctly identified. Specificity, on the other hand, measures the percentage of counties where Trump did not receive a majority that the model correctly identified. Finally, precision measures the percentage of the model's positive predictions (Trump receiving more than 50%) that were actually correct. In this dataset, all three models have higher specificity than sensitivity, which means they are better at identifying the counties where Trump did not get the majority than the counties where he did.

After the examination of each method, the performance of each method is compared with each other. Figure14 presents the boxplots of the predictive accuracy across the 10 folds for each method. Random Forest consistently achieves the highest accuracy with the smallest variance, while logistic regression shows more variability. SVM falls between the two, with a distribution centered around 75%.

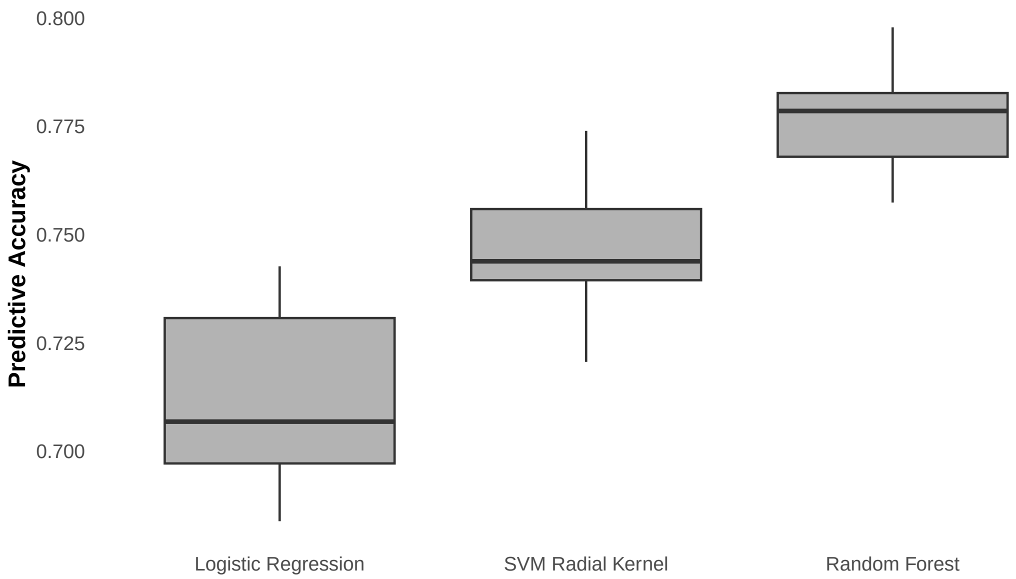

Figure 14: Boxplots of predictive accuracy across 10 folds for each classification method

Table2 summarizes the classification performance of the three methods across the 2-times repeated stratified 5-fold cross-validation.

Table 2: Classification performance across 10 folds

Method	Mean Accuracy	SD	Sensitivity	Specificity	Precision
Random Forest	0.776	0.012	0.561	0.901	0.768
SVM	0.748	0.016	0.477	0.904	0.744
Logistic Regression	0.713	0.020	0.454	0.862	0.658

Random Forest was the best performing method, achieving the highest accuracy, sensitivity, and precision, while also being the most stable across folds. SVM performed second best and had the highest specificity (90.4%), meaning it was the best model correctly identifying counties where Trump did not receive a majority across the other Republican candidates. On the other hand, Logistic regression performed the lowest but has the advantage of interpretability, as the backward stepwise procedure identified a stable set of important predictors.

An important observation is that all three methods performed better than the majority class baseline, which would give an accuracy of 63.4% by always predicting that Trump did not receive a majority. This confirms that the socioeconomic characteristics of the counties carry useful predictive information and also the demographic and economic profile of a county is strongly linked to the Republican primary voting patterns. At the same time, the remaining 22% of misclassified counties indicates that there are additional factors beyond socioeconomic characteristics that influence the outcome.

Across all three methods, the variables that emerged as the most consistently important were the racial composition of the county (particularly the percentage of Hispanic and Black residents), education (bachelor's degree percentage), age distribution (percentage under 18), and land area. These findings indicate that Trump tended to receive more than 50% of the Republican primary votes in counties that were less diverse, less educated at the college level, had older populations, and covered larger land areas.

The ROC curve and the Area Under the Curve (AUC) were not used for model comparison because they require predicted probabilities from each model. While logistic regression provides probabilities directly, Support Vector Machines is a hard classifier, assigning 0 and 1 to classes. Also, the aim is predicting at the specific threshold of 50%, though ROC curve is used accessing binary classifiers' performance for thresholds from 0% to 100%, making it unuseful to be calculated.

After identifying Random Forest as the best performing method, if the goal were to deploy this model for predicting Trump's majority in new counties, the final step would be to refit the Random Forest model on the entire dataset of 2,721 counties. The cross-validation was used only to estimate how well the model would perform on unseen data, but the final model should be trained on all available data to make the most of the information available.

6 Clustering Analysis: Grouping Counties by Demographics

The second part of the analysis focuses on clustering the US counties based on the 22 demographic variables only, without using any information about the votes. The goal is to identify natural clusters of counties that share similar demographic characteristics and then describe these groups using the 29 economic variables. As described in Section 2, the clustering uses all 3,143 counties, not only the 2,721 that were merged with the voting data. The clustering was performed using model-based clustering which fits finite mixtures of Gaussian distributions using the EM algorithm, while best model is selected based on the Bayesian Information Criterion (BIC).

Variable Selection

As shown in the correlation analysis in Section 4, several of these demographic variables carry similar information. The three population count variables (population 2014, population 2010 estimates base, and population 2010 census) were all associated with each other with $r = 1.00$, meaning they measure exactly the same thing. The number of veterans was also nearly perfectly correlated with population ($r = .98$), indicating that it is simply an absolute count that reflects county size rather than providing information about the share of veterans. Therefore, three of the four population counts and the veterans variable were removed, keeping only the 2014 population estimate. Additionally, the percentage of persons under 5 years was highly correlated with the percentage of persons under 18 ($r = .83$), which is expected because persons under 5 are also persons under 18, so the variable under 5 was removed. Moreover, the percentage of White alone was strongly associated with the percentage of White alone not Hispanic ($r = .85$), and since the second one is more specific, we removed the White alone variable. Finally, the percentage of people who speak a language other than English at home was dropped, as it was correlated with the percentage of foreign-born persons ($r = .86$).

Next, Table 3 summarizes the variables that were dropped and the reason for each as discussed above.

Table 3: Demographic variables removed due to high correlations

Dropped Variable	Kept Variable	r	Reason
Population 2010 (est. base)	Population 2014	1.00	Identical measure
Population 2010 (census)	Population 2014	1.00	Identical measure
Veterans (count)	Population 2014	.98	Absolute count, reflects size
Persons under 5 (%)	Persons under 18 (%)	.83	Subset of under 18
White alone (%)	White not Hispanic (%)	.85	Less specific measure
Non-English at home (%)	Foreign born (%)	.86	Captures same concept

After the dropping of the correlated variables, Model-based clustering was then applied to the 16 remaining demographic variables, with the option of two to five clusters. The Bayesian Information Criterion selected a model of five ellipsoidal clusters with varying volume, shape, and orientation, and in order to check whether all 16 variables contribute to the clustering, the cluster means were examined. The percentage of female persons (49.1–50.7%), the percentage of Native Hawaiian residents (0.0–0.9%), the percentage of persons identifying as two or more races (1.2–4.6%), and the percentage of persons living in the same house (84.3–88.5%) were found not varying between the different clusters.

Subsequently, in order to confirm that these variables should be removed, the Bayesian Information Criterion of the model was compared before and after dropping each variable. The variables were tested one after the other, and not all at once, because removing one variable can change how important the remaining ones are. Each time a variable was removed and the BIC got better, the variable was dropped and the next one was tested on the smaller set.

The percentage of female persons was removed first, improving the BIC by 10,930, followed by the percentage of two or more races. Next, the percentage of persons living in the same house was dropped from the 14-variable model, improving the BIC by another 16,649. Finally, the percentage of Native Hawaiian residents was tested on the model with 13 variables, but removing it made the BIC worse by 5,076, so it was kept. After this procedure, 13 demographic variables remained for the final clustering.

Model-Based Clustering

Next, three different models were fitted to the 13 variables with the number of clusters set to three, four and five, without standardizing the data, since the model estimates separate covariance matrices for each cluster and can handle different scales. Table 4 compares the BIC and ICL values for the three solutions.

Table 4: BIC and ICL comparison for different numbers of clusters

k	BIC	ICL	BIC improvement
3	−243,700	−243,834	–
4	−235,724	−235,945	+7,976
5	−231,010	−231,285	+4,714

Both BIC and ICL improved from $k = 3$ to $k = 5$, with the largest improvement occurring from $k = 3$ to $k = 4$ (BIC improved by 7,976) and a smaller improvement from $k = 4$ to $k = 5$ (BIC improved by 4,714). However, the solution with four clusters provides more interpretable county groups, and thus $k = 4$ was selected.

In addition, backward variable selection was performed, attempting to remove each variable and none could be dropped without worsening the model fit, which confirms that all 13 variables contribute to the clustering.

The final model has log-likelihood = −116,175, BIC = −235,724, and ICL = −235,945. Moreover, the average classification uncertainty across all 3,143 counties was 3.0%, indicating that most counties were assigned to the correct cluster. Only 184 counties out of 3,143 had uncertainty above 20%, representing countries cases between two clusters.

Table 5 presents the mean values of the 13 demographic variables for each of the four clusters.

Table 5: Cluster profiles: mean values of the 13 demographic variables by cluster

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Population (2014)	148,994	446,552	17,388	44,824
Population change (%)	2.3	3.2	−0.8	−0.7
Persons under 18 (%)	23.1	24.3	22.0	22.0
Persons 65+ (%)	15.9	14.3	20.6	17.4
Black (%)	9.4	7.1	0.7	16.1
American Indian (%)	1.2	15.7	1.7	0.4
Asian (%)	2.1	5.0	0.5	0.7
Native Hawaiian (%)	0.1	0.7	0.1	0.0
Hispanic (%)	19.3	14.3	3.9	3.1
White not Hispanic (%)	67.6	54.9	92.1	78.8
Foreign born (%)	8.3	10.0	1.9	2.0
High school grad. (%)	84.1	85.3	88.9	81.4
Bachelor's degree (%)	24.1	23.5	18.9	15.9
Cluster size (n)	924	261	843	1,115

Table 5 presents the mean values of the 13 demographic variables for each of the four clusters. What separates the four clusters is mainly the racial composition, the population size, and the age structure. Clusters 1 and 2 are the growing and diverse counties, but they differ in size. Cluster 2 contains the large metropolitan areas with almost half a million residents on average, while Cluster 1 has mid-sized counties around 149,000. Furthermore, Cluster 2 also has a very high American Indian percentage (15.7%), which is much higher than any other cluster. Both clusters have high shares of foreign-born residents and Hispanic populations, and higher education levels compared to the other two groups.

On the other side, Clusters 3 and 4 are both losing population, but they look very different from each other. Cluster 3 is almost entirely White (92.1%) with very few minorities or immigrants. These are the oldest counties, with more 20% residents aged 65 and over, and they have the highest high school graduation rate but fewer people with a university degree. Cluster 4 has the highest Black population (16.1%) and the lowest education levels across both high school (81.4%) and bachelor's degree (15.9%).

So the main split in the data is between growing diverse counties (Clusters 1 and 2) and declining less diverse counties (Clusters 3 and 4), with a secondary split based on whether the county is White with an aging population (Cluster 3) or has a large Black population with low education (Cluster 4).

Cluster Validation

After the identification four different clusters, the dataset was randomly divided into two halves (1,571 and 1,572 counties), and two models with four clusters were fitted separately to each half. Table 6 and Table 7 present the cluster profiles for the two samples.

Table 6: Cluster profiles for sample A

Variable	Cluster A1	Cluster A2	Cluster A3	Cluster A4
Population (mean)	136,937	18,192	51,274	536,420
Black (%)	8.4	0.7	19.0	6.5
Hispanic (%)	18.7	2.7	3.8	15.4
White not Hispanic (%)	68.5	94.4	75.0	52.9
Foreign born (%)	7.8	1.5	2.5	11.2
Bachelor's degree (%)	23.5	17.5	16.8	23.0
Cluster size (n)	499	420	522	130

Table 7: Cluster profiles for sample B

Variable	Cluster B1	Cluster B2	Cluster B3	Cluster B4
Population (mean)	18,558	91,794	65,649	459,103
Black (%)	0.9	4.9	19.4	11.1
Hispanic (%)	2.9	19.0	3.9	12.7
White not Hispanic (%)	94.1	71.5	74.2	54.7
Foreign born (%)	1.5	7.3	2.8	10.7
Bachelor's degree (%)	16.8	22.1	18.9	26.5
Cluster size (n)	483	460	492	137

Although the cluster labels differ between the two halves (because the numbering is arbitrary), the same four county types emerge in both samples. In sample A, cluster A2 contains the predominantly White counties (94.4% White), cluster A3 contains the counties with high Black population (19.0%), cluster A1 contains the growing diverse counties with high Hispanic population (18.7%), and cluster A4 contains the large metropolitan counties (highest population and foreign-born percentage). As can be seen for the Table above, the four cluster are also appearing in the sample B, which confirms that the clusters found are actually natural groups of the counties.

Cluster Interpretation using Economic Variables

After identifying the four demographic clusters, the economic variables were used to describe the economic conditions of each group.

The most economically distinct group of counties is Cluster 2, which has the highest median household income ($51,282), the highest housing values ($180,524), and the most building permits (M = 1,518), reflecting the economic activity of large urban areas. At the same time, this cluster has the lowest homeownership rate (65.2%) and the highest multi-unit housing (18.8%), which is typical of cities where renting is more common than owning. These economic characteristics make sense given the demographic profile of Cluster 2, because these large metropolitan counties attract immigrants, likely because of job opportunities and high-paying jobs. The high incomes and high housing values reflect the nature of job opportunities, while the low homeownership and high multi-unit housing are a natural result of expensive urban housing market where renting is more affordable than buying a home.

On the opposite end, Cluster 3 has the highest homeownership rate (76.3%) and the lowest poverty rate (13.5%), but moderate housing values ($117,090) and the lowest population density (26.5 per square mile). These are affordable places where most people own their homes, but with limited economic growth, as shown by the low number of building permits (M = 38.5). The low poverty rate may seem surprising given the moderate incomes ($46,171), but it could be explained by the low cost of living in these areas. With over 92% White population and very few immigrants, these counties may be based on economies, such as agricalture.

Cluster 4 is characterized by the worst economic conditions, as it has the lowest median household income ($41,087), the lowest per capita income ($21,174), and the highest poverty rate (19.3%). The commute times are also the longest (M = 25.3 minutes), which indicate that jobs are not located near where people live, while residents also receive the lowest compensation across the groups. Furthermore, the low education levels (81.4% high school graduation, the lowest among the four clusters) may further limit access to better paying jobs.

Lastly, income levels ($50,067) in Cluster 1 is closer to Cluster 2 than to Clusters 3 and 4, and it has moderate homeownership (69.3%) and moderate density (363 per square mile). The mean travel time to work is the shortest (M = 22.2 minutes) and the housing values are the second highest ($156,344). Additionally, the short commute times could suggest that jobs are more accessible in these areas compared to the rural White counties or the high-Black counties.

An interesting insight is that poverty does not simply follow income, as Cluster 2 has the highest incomes but also a high poverty rate (18.1%), almost equal to Cluster 4 (19.3%) which has the lowest incomes. This may be because large urban counties have both rich and poor areas or maybe because many of the immigrants who move to these counties for work may earn low wages when they first arrive, which pushes the poverty rate up even though the average income of the county is high.

Table 8 and Figures 15 and 16 present the economic characteristics of the four clusters.

Table 8: Economic characteristics by cluster ($N = 3{,}143$).

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Median household income ($)	50,067	51,282	46,171	41,087
Per capita income ($)	25,349	25,203	24,489	21,174
Persons below poverty (%)	16.1	18.1	13.5	19.3
Homeownership rate (%)	69.3	65.2	76.3	73.0
Median housing value ($)	156,344	180,524	117,090	108,720
Multi-unit housing (%)	16.1	18.8	9.0	10.4
Mean travel time to work (min)	22.2	21.9	21.1	25.3
Retail sales per capita ($)	11,572	10,882	9,824	9,290
Women-owned firms (%)	20.4	20.8	12.6	19.3
Building permits	554	1,518	39	95
Population per sq. mile	363	1,270	27	113
Land area (sq. miles)	1,117	3,848	1,083	522

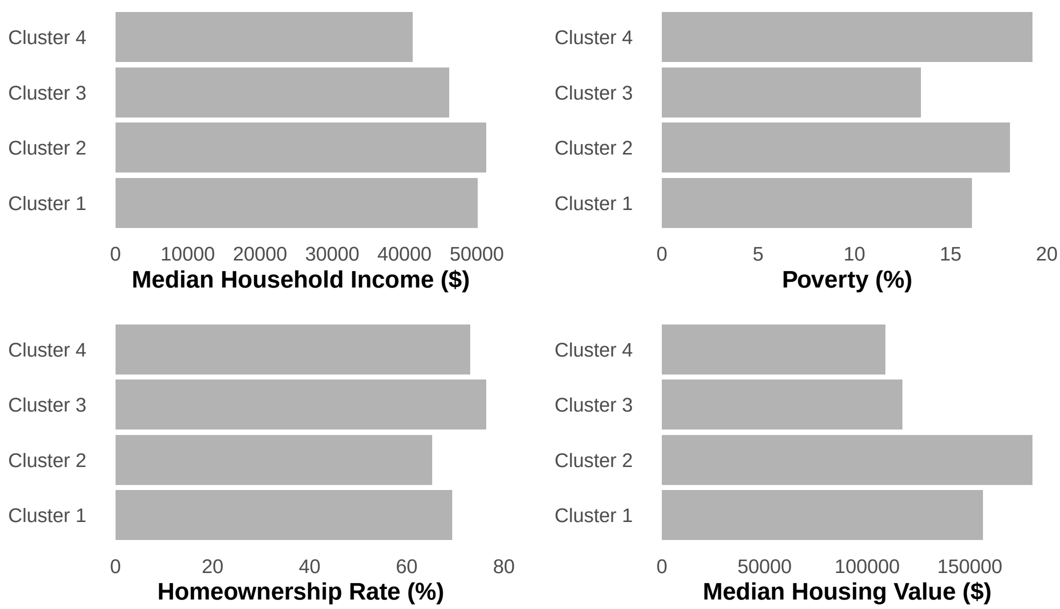

Figure 15: Mean economic characteristics by cluster (a) Median household income (b) Poverty rate (c) Homeownership rate (d) Median housing value.

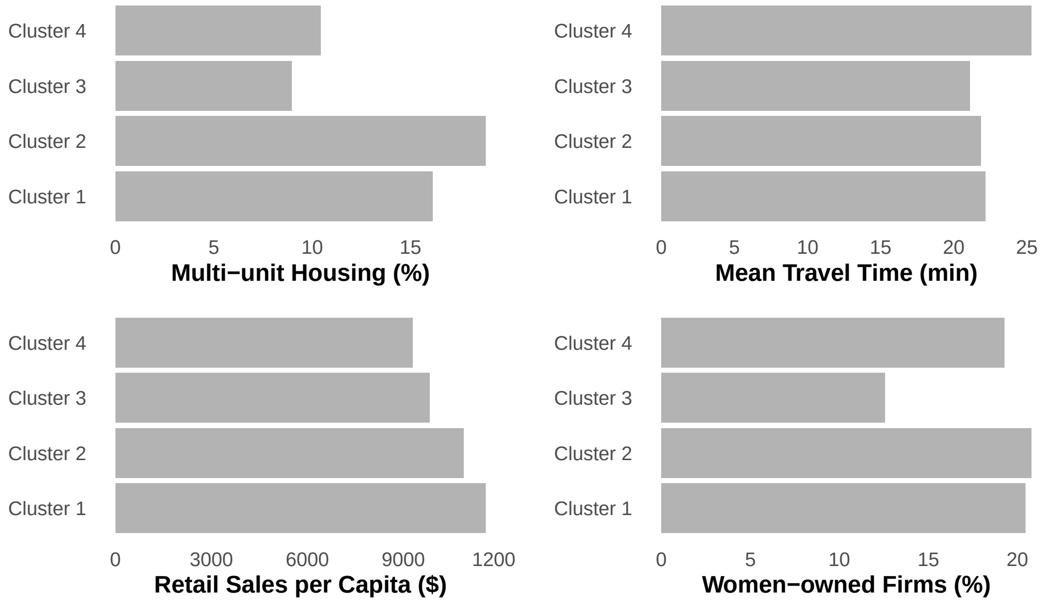

Figure 16: Mean economic characteristics by cluster (a) Multi-unit housing (b) Mean travel time to work (c) Retail sales per capita (d) Women-owned firms.

7 Conclusion and Connection Between Classification and Clustering

This report analyzed data from the 2016 US presidential primaries at the county level, with the goal of understanding the relationship between the socioeconomic characteristics of US counties and the votes. The first part built classification models to predict whether Trump received more than 50% of the Republican primary votes in each county and the second part clustered the counties based on their demographic characteristics and described the resulting groups using the economic variables.

In the classification part, the three models of Logistic Regression, Support Vector Machines with a radial kernel, and Random Forest were built. All models performed better than a model just predicting the majority class of 63.4%, while Random Forest achieved the highest accuracy of 77.6%. The variables that were consistently important in Logistic Regression and Random Forest were the racial composition of the county (particularly the percentage of Hispanic and Black residents), land area, median housing value, and the age distribution. SVM does not provide variable importance measures, but its strong performance suggests that similar patterns drive its predictions. These findings suggest that the demographic and economic profile of a county carries meaningful predictive information about the Republican primary votes.

In the clustering part, model-based clustering identified four distinct groups of US counties based on 13 demographic variables. The first group ($n = 924$) contains growing counties with a high Hispanic population (19.3%) and moderate diversity, where incomes are relatively high

(\$50,067) and commute times are the shortest (22.2 minutes). In turn, the second group (n = 261) includes the large metropolitan counties with the highest foreign-born (10.0%) and Asian (5.0%) populations, where incomes and housing values are the highest, but homeownership is the lowest (65.2%) and poverty remains high (18.1%) due to the mix of wealthy and poor districts. The third cluster (n = 843) consists of mostly White (92.1%), aging, and declining counties, where housing is affordable, homeownership is the highest (76.3%), and poverty is the lowest (13.5%), but economic growth is limited with very few building permits. The fourth and final group (n = 1,115) contains counties with the highest Black population (16.1%), where economic conditions are the hardest, with the lowest incomes (\$41,087), the highest poverty (19.3%), the lowest education levels, and the longest commute times (25.3 minutes).

Although the clustering was performed independently of the voting data, the four clusters show different voting patterns. Table 9 presents the percentage of counties in each cluster where Trump received more than 50% of the Republican primary votes, as well as his mean fraction of votes.

Table 9: Trump majority rate and mean fraction of votes by cluster.

Cluster	Trump > 50%	Mean Fraction
Cluster 1 (high Hispanic, $n = 924$)	27.7%	0.423
Cluster 2 (urban diverse, $n = 261$)	47.3%	0.507
Cluster 3 (White rural, $n = 843$)	47.4%	0.505
Cluster 4 (high Black, $n = 1{,}115$)	34.9%	0.475

Clusters 2 and 3 had the highest rates of Trump majority (47.3% and 47.4%) and the highest mean fraction of votes (0.507 and 0.505), in contract with Cluster 1 and 4, in which Trump had the lowest rates of 27.7% and 34.9% respectively. As a result, the White rural counties (Cluster 3) and the large urban diverse counties (Cluster 2) both gave Trump high support, even though they are very different demographically. One possible explanation is that in the Republican primaries, Trump appealed to both rural White voters and voters in large diverse metropolitan areas, but for different reasons. In comparison, the counties with high Hispanic populations (Cluster 1) gave the least votes to Trump, which is consistent with the classification results where Hispanic percentage was one of the most important predictors.

Additional data that could improve this analysis include the urban or rural classification of each county, the presence or not of universities, and income inequality measures such as the Gini coefficient. These variables could help maybe explain some of the variation within the clusters and improve the predictive accuracy of the classification models.